

Chapter 5

Fisher's z -distribution

5.1 Fisher's z -distribution

We have, from Snedecor's F -distribution, as

$$f(F) = \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \frac{F^{\frac{n_1}{2}-1}}{\left(1 + \frac{n_1}{n_2}F\right)^{\frac{n_1+n_2}{2}}}, \quad 0 < F < \infty.$$

Here take

$$F = e^{2Z} \Rightarrow \frac{dF}{dz} = 2e^{2z}, \quad \text{and } Z = \frac{1}{2} \log_e F.$$

The distribution of Z becomes

$$\begin{aligned} g(z) &= f(F) \cdot \left| \frac{dF}{dz} \right| \\ &= \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \frac{(e^{2z})^{\frac{n_1}{2}-1}}{\left(1 + \frac{n_1}{n_2}e^{2z}\right)^{\frac{n_1+n_2}{2}}} \cdot 2e^{2z} \\ &= 2 \cdot \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \frac{e^{n_1 z}}{\left(1 + \frac{n_1}{n_2}e^{2z}\right)^{\frac{n_1+n_2}{2}}}, \quad -\infty < z < \infty, \end{aligned}$$

which is the density function of Fisher's z -distribution with (n_1, n_2) d.f.

5.2 Moment Generating Function of z -distribution

The moment generating function of z -distribution can be obtain as

$$\begin{aligned}
 M_Z(t) &= E(e^{tZ}) \\
 &= \int_{-\infty}^{\infty} e^{tz} g(z) dz \\
 &= \int_{-\infty}^{\infty} e^{tz} \left[2 \cdot \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \frac{e^{n_1 z}}{\left(1 + \frac{n_1}{n_2} e^{2z}\right)^{\frac{n_1+n_2}{2}}} \right] dz \\
 &= \int_{-\infty}^{\infty} (e^{2z})^{\frac{t}{2}} \left[2 \cdot \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \frac{(e^{2z})^{\frac{n_1}{2}}}{\left(1 + \frac{n_1}{n_2} e^{2z}\right)^{\frac{n_1+n_2}{2}}} \right] dz. \tag{5.1}
 \end{aligned}$$

Set $e^{2z} = F$ then $\frac{dF}{dz} = 2e^{2z} \Rightarrow dz = \frac{1}{2e^{2z}} dF$. Using (5.1), we have

$$\begin{aligned}
 M_Z(t) &= \int_0^{\infty} F^{\frac{t}{2}} \left[2 \cdot \frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \frac{F^{\frac{n_1}{2}}}{\left(1 + \frac{n_1}{n_2} F\right)^{\frac{n_1+n_2}{2}}} \right] \frac{1}{2e^{2z}} dF \\
 &= \int_0^{\infty} F^{\frac{t}{2}} \left[\frac{\left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}}}{B\left(\frac{n_1}{2}, \frac{n_2}{2}\right)} \cdot \frac{F^{\frac{n_1}{2}-1}}{\left(1 + \frac{n_1}{n_2} F\right)^{\frac{n_1+n_2}{2}}} \right] dF \\
 &= \int_0^{\infty} F^{\frac{t}{2}} f(F) dF.
 \end{aligned}$$

Since μ'_r for F -distribution is $\int_0^{\infty} F^r f(F) dF$, hence we can find moment generating function of the z -distribution by setting $r = \frac{t}{2}$ in above expression and can be expressed as follow

$$M_Z(t) = \left(\frac{n_2}{n_1}\right)^{\frac{t}{2}} \frac{\Gamma\left(\frac{n_1}{2} + \frac{t}{2}\right) \Gamma\left(\frac{n_2}{2} - \frac{t}{2}\right)}{\Gamma\left(\frac{n_1}{2}\right) \Gamma\left(\frac{n_2}{2}\right)}, \quad n_2 > t.$$

5.3 Fisher's Z -transformation

This transformation is suggested for stabilizing the variance of the distribution of correlation coefficient. If r is the sample correlation coefficient in sampling from

a correlated bivariate normal population with correlation coefficient ρ then the asymptotic distribution of r , as $n \rightarrow \infty$ is normal with mean

$$E(r) = \rho,$$

and

$$Var(r) = \frac{(1 - \rho^2)^2}{n}.$$

In the usual notations we have

$$\Psi(\rho) = \frac{(1 - \rho^2)^2}{n}.$$

Using $g(\theta) = \int \frac{c}{\sqrt{\Psi(\theta)}} d\theta$, the transformation function is

$$\begin{aligned} g(\rho) &= \int \frac{c\sqrt{n}}{1 - \rho^2} d\rho \\ &= c\sqrt{n} \cdot \frac{1}{2} \log_e \left(\frac{1 + \rho}{1 - \rho} \right). \end{aligned}$$

We select c in such a way that

$$c\sqrt{n} = 1 \quad \Rightarrow c = \frac{1}{\sqrt{n}},$$

so that

$$g(\rho) = \frac{1}{2} \cdot \log_e \left(\frac{1 + \rho}{1 - \rho} \right).$$

Hence the transformation for the statistic is

$$g(r) = \frac{1}{2} \cdot \log_e \left(\frac{1 + r}{1 - r} \right),$$

which is denoted by Z , and using $g(T) = N [g(\theta), [g'(\theta)]^2 \cdot \Psi(\theta)]$, the transformed statistic is normally distributed with mean

$$g(\rho) = \frac{1}{2} \cdot \log_e \left(\frac{1+\rho}{1-\rho} \right),$$

and variance

$$\begin{aligned} \text{Var} [g(r)] &= [g'(\rho)]^2 \cdot \Psi(\rho) \\ &= \left(\frac{1}{1-\rho^2} \right)^2 \cdot \frac{(1-\rho^2)^2}{n} \\ &= \frac{1}{n}, \text{ for large } n. \end{aligned}$$

Or alternatively

$$\text{Var} [g(r)] = c^2 = \left(\frac{1}{\sqrt{n}} \right)^2 = \frac{1}{n}.$$

Hence $Z = \frac{1}{2} \cdot \log_e \left(\frac{1+r}{1-r} \right) \sim N \left(\frac{1}{2} \cdot \log_e \left(\frac{1+\rho}{1-\rho} \right), \frac{1}{n} \right)$, for large n .

5.4 Applications of Fisher's z -transformation

To: test the significant difference between sample and population correlation coefficient, test the significance of the difference between two independent sample correlation coefficients, obtain pooled estimate of population correlation coefficient.